MOTOR-PUMP FIELD INSPECTION REPORT

MotorGuard Digital Twin — Predictive Maintenance System v2.1

Report No.	RPT-2026-4379
Date & Time	Saturday, 23 May 2026 at 11:37 pm
Machine ID	MOTOR-3D-TWIN
Inspector	System Administrator
Site	Takoradi
Work Order	WO-2026. 11

1 JOB INFORMATION

Work Order / PM Ticket	WO-2026. 11	Inspection Type	Predictive
Site / Location	Takoradi	Date of Inspection	23/05/2026
Shift	Day	Department / Team	–
Inspector Name	System Administrator	Contact No.	–

2 EQUIPMENT NAMEPLATE DATA

MOTOR

Manufacturer	—
Model / Type	Squirrel-Cage Induction Motor (SCIM)
Serial No.	—
Frame Size (IEC)	—
Rated Power	15 kW
Voltage / Supply	400 V / 3Ø / 50 Hz
Rated Current	30 A
Rated Speed	1480 RPM
Efficiency Class	IE3
Insulation Class	Class F
Duty Cycle	S1
Thermal Protect.	Thermistor (PTC)
Enclosure (IP)	IP55 / TEFC
Mounting	B3 (Foot)
Installation Date	—
Operating Hours	—
Rewound	No

PUMP

Manufacturer	—
Model / Type	End-Suction Centrifugal Pump
Serial No.	—
Flow Rate	45 m^3/h
Total Head	32 m
Rated Speed	1480 RPM
Impeller Dia.	210 mm
Seal Type	Mechanical seal
Orientation	Horizontal
Installation Date	—

3 PRE-START & STARTUP CHECKLIST

Completed before motor start. Values entered by inspector on site.

ITEM	REQUIREMENT	OBSERVED VALUE	STATUS
Supply Voltage L1–L2	400 V ±10% (360–440 V)	–	—
Supply Voltage L2–L3	400 V ±10% (360–440 V)	–	—
Supply Voltage L1–L3	400 V ±10% (360–440 V)	–	—
Supply Frequency	50 Hz ±2% (49–51 Hz)	–	—
Insulation Resistance (Megger)	≥1 MΩ at 500 V DC	–	—
Earth Continuity	<1 Ω	–	—
Bearing Lubrication	Per manufacturer spec	OK	PASS
Cooling / Ventilation	Clear, unobstructed, operational	OK	PASS
Guards & Safety Covers	All in place and secure	OK	PASS
Coupling Alignment	Within manufacturer tolerance	OK	PASS
Motor Cleanliness	Free from dust / oil accumulation	OK	PASS
Mounting Bolts Torqued	Per torque spec	Yes	PASS

4 LIVE ELECTRICAL PARAMETERS — RUNNING CONDITION

4.1 — THREE-PHASE SUPPLY

PARAMETER	READING	RATED / LIMIT	STATUS
Supply Voltage	— V	Rated: 400 V | 360–440 V range	—
Supply Frequency	— Hz	Rated: 50 Hz | 49–51 Hz range	—
Power Factor	—	Min: 0.85 (lagging)	—
Active Power	— kW	Rated: 15 kW	—
Apparent Power	— kVA		—
THD — Current	— %	IEC 61000-3-2 | Limit: <5%	—

4.2 — STATOR PHASE CURRENTS (IEC 60034-1)

PHASE	CURRENT (A)	RATED (A)	DEVIATION FROM MEAN	STATUS
L1 (U)	102.88	30 A	0.0 %	NORMAL
L2 (V)	102.83	30 A	-0.0 %	NORMAL
L3 (W)	102.92	30 A	0.0 %	NORMAL
Phase Imbalance	**0.1 %**		Warning: >8% | Critical: >12%	NORMAL

NEMA MG-1 / IEC 60034-26: Sustained phase imbalance >1% causes additional rotor heating. Values >5% risk premature failure.

5 LIVE MECHANICAL PARAMETERS — RUNNING CONDITION

5.1 — SPEED & LOAD

PARAMETER	READING	RATED / LIMIT	STATUS
Rotor Speed	1500 RPM	Rated: 1480 | Warn: >1520 | Crit: >1550	NORMAL
Shaft Torque	7.5 N·m	Rated: 97.3 N·m | Warn: >90 | Crit: >110	NORMAL
Load Factor	– %	Normal: ≤100% | Warn: >90%	—

5.2 — VIBRATION (ISO 10816-3)

PARAMETER	READING	RATED / LIMIT	STATUS
Vibration RMS	0.000 g	Zone A: <2.3 g | Zone C: <7.1 g | Crit: ≥8.0 g	NORMAL
Vibration Kurtosis	0.00	Normal: <4 | Warn: ≥10 | Crit: ≥16	NORMAL
Crest Factor	1.00	Normal: <4 | Warn: ≥6 | Crit: ≥8	NORMAL

ISO 10816-3: Zone A=new machines | Zone B=acceptable long-term | Zone C=alarm, investigate | Zone D=danger, stop machine.

6 LIVE THERMAL PARAMETERS (IEC 60034-1)

PARAMETER	READING	RATED / LIMIT	STATUS
Stator Winding Temp.	300.0 °C	Warn: 65 °C | Crit: 85 °C | Class F abs. limit: 155 °C	CRITICAL
Bearing Temp. (DE)	295.0 °C	Warn: 70 °C | Crit: 85 °C	CRITICAL
Ambient Temperature	1.0 °C	Max: 40 °C (IEC 60034-1 standard rating)	NORMAL
Temperature Rise (ΔT)	299.0 K	Warn: 20 K | Crit: 30 K (Stator – Ambient)	CRITICAL
Thermal Hotspot (IR)	300.0 °C	Warn: 80 °C | Crit: 95 °C	CRITICAL

IEC 60034-1: Each 10 °C above rated temperature halves insulation life (Arrhenius rule). Maintain adequate ventilation at all times.

7 AI PREDICTIVE DIAGNOSTIC RESULT

Model: Meta-Fusion (CWRU-CNN + Induction-CNN + NASA Bi-LSTM-Attn + Current-CNN + Thermal-MobileNetV2) · System accuracy: 90.67 % · F1 macro: 0.906

HEALTH CLASSIFICATION	REMAINING USEFUL LIFE	PREDICTION CERTAINTY	MODEL ACCURACY
■ CRITICAL	35 h (≈ 4 working days)	82% Per this inference	90.67 % F1 = 0.906

0 h — Critical | 150 h — Warning | 500 h — Full life

Fault Status	CRITICAL
Interpretation	*Critical parameter exceedance detected. Suspend operation, implement lockout/tagout, and notify the site engineer immediately before any restart.*
Required Action	**Stop motor immediately if safe. Lockout/tagout. Notify maintenance engineer before restart.**

AI results are based on sensor data from MATLAB/Simulink via MotorGuard Digital Twin v2.1. Always verify AI results against physical inspection before acting. For field use only.

8 MAINTENANCE ACTIONS PERFORMED

WORK CARRIED OUT (DESCRIBE ALL ACTIONS TAKEN)

(No entries)

PARTS / MATERIALS REPLACED

PART / COMPONENT	PART NUMBER	QUANTITY	SUPPLIER
No parts replaced			

NEXT MAINTENANCE

TYPE	DUE DATE
Routine	DD/MM/YYYY

9 ACTIVE ALARMS AT TIME OF INSPECTION

SEVERIT Y	DESCRIPTION	SOURCE	TIME RAISED	AC K.	REQUIRED ACTION
CRITIC AL	Health state escalated to critical	Fusion Engine	11:37:39 pm	Yes	*Notify engineer — isolate machine*
CRITIC AL	Stator temperature exceeded limit (300.0 C)	Thermal	11:37:39 pm	Yes	*Notify engineer — isolate machine*
CRITIC AL	RUL below threshold — 35 h remaining	NASA/RUL	11:37:39 pm	Yes	*Notify engineer — isolate machine*
WARNI NG	Data stream is stale	Transport	11:37:39 pm	Yes	*Investigate within 10 min*

10 INSPECTION RECORD & SIGN-OFF

INSPECTOR NAME
System Administrator

DESIGNATION / ROLE
operator

INSPECTOR SIGNATURE
____

DATE OF INSPECTION
23/05/2026

SUPERVISOR NAME
Silas Asare

SUPERVISOR SIGNATURE
____

SUPERVISOR DATE
____

NEXT INSPECTION DUE
23/052026

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